UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

**CHEMISTRY** **5070/21**

Paper 2 Theory **May/June 2013**

**1 hour 30 minutes**

Candidates answer on the Question Paper.

No Additional Materials are required.

**READ THESE INSTRUCTIONS FIRST**

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

**Section A**
Answer **all** questions.
Write your answers in the spaces provided in the Question Paper.

**Section B**
Answer any **three** questions.
Write your answers in the spaces provided in the Question Paper.

Electronic calculators may be used.

A copy of the Periodic Table is printed on page 20.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

This document consists of **17** printed pages and **3** blank pages.


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## Section A

*For Examiner's Use*

Answer **all** the questions in this section in the spaces provided.

The total mark for this section is 45.

**A1** Choose from the following compounds to answer the questions below.

**butane**
**calcium carbonate**
**carbon dioxide**
**copper(II) nitrate**
**iron(II) hydroxide**
**iron(III) hydroxide**
**propene**
**sodium chloride**
**sulfur dioxide**
**sulfuric acid**

Each compound can be used once, more than once or not at all.

Name a compound which

**(a)** is a green solid,

..... [1]

**(b)** is a saturated hydrocarbon,

..... [1]

**(c)** has a molecule with only 9 atoms,

..... [1]

**(d)** can be used to reduce the acidity in lakes,

..... [1]

**(e)** will turn aqueous acidified potassium dichromate(VI) from orange to green,

..... [1]

**(f)** can be electrolysed in aqueous solution to form two gases.

..... [1]

[Total: 6]

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**A2** Photosynthesis helps to maintain the percentage of oxygen in air.

*For Examiner's Use*

**(a)** What is the percentage, by volume, of oxygen in dry air?

..... [1]

**(b)** In addition to releasing oxygen, photosynthesis produces glucose, $C_6H_{12}O_6$.

Write the overall equation that represents photosynthesis.

..... [1]

**(c)** Describe the essential conditions needed for photosynthesis.

.....

.....

..... [2]

**(d)** Photosynthesis is an endothermic reaction.

**(i)** Explain, in terms of the energy changes that occur during bond breaking and bond making, why photosynthesis is an endothermic reaction.

.....

.....

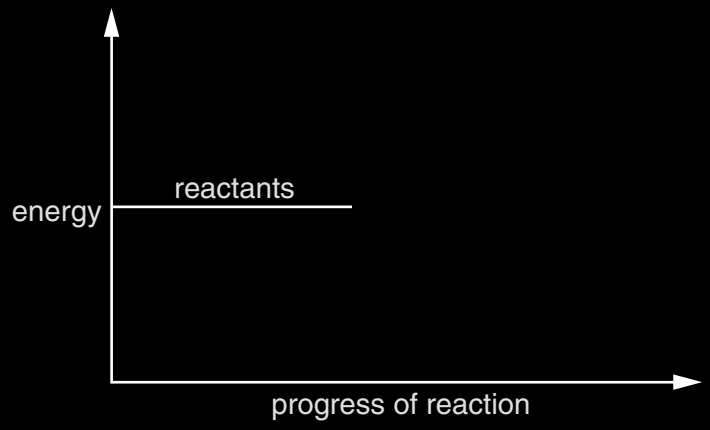
.....

..... [2]

**(ii)** Complete the energy profile diagram for photosynthesis.

On your diagram label the

- products,
- enthalpy change for the reaction, $\Delta H$,
- activation energy, $E_a$.




[3]

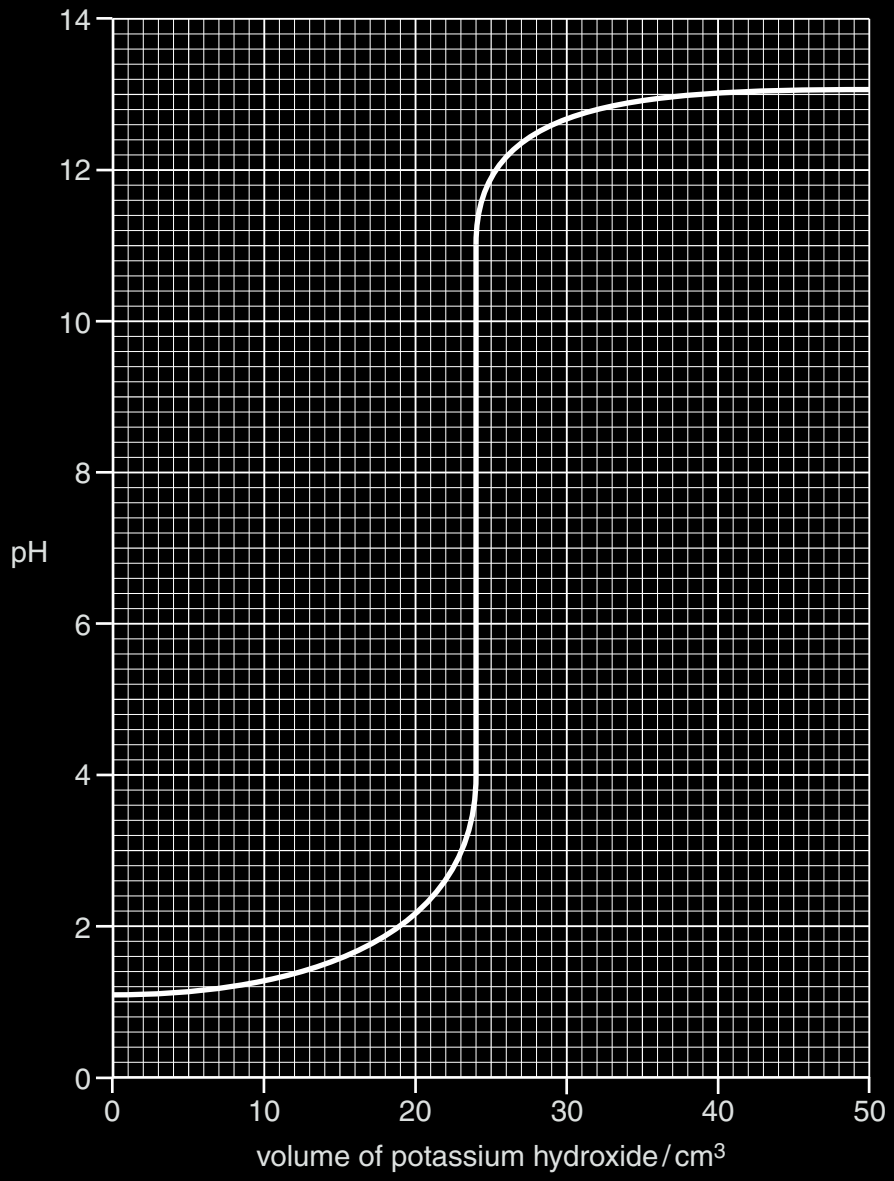
[Total: 9]

**A3** Salts are often made by the neutralisation of bases.

*For Examiner's Use*

**(a)** Aqueous potassium hydroxide, of concentration 0.150 mol / $dm^3$, is added to 25.0 $cm^3$ of sulfuric acid in a flask.

The graph shows how the pH of the liquid in the flask changes as aqueous potassium hydroxide is added to it.




**(i)** Construct the equation for the complete neutralisation of sulfuric acid by potassium hydroxide.

..... [1]

**(ii)** Use the graph to deduce the volume of aqueous potassium hydroxide required to neutralise 25.0 $cm^3$ of sulfuric acid.

..... [1]

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**(iii)** Use your answers to **(i)** and **(ii)** to calculate the concentration of sulfuric acid.

*For Examiner's Use*

concentration of sulfuric acid = ..... $mol/dm^3$ [3]

**(b)** Describe the essential experimental details for preparing a pure sample of zinc nitrate crystals from zinc oxide.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

[Total: 9]

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**A4** The table shows the number of electrons, neutrons and protons in seven different particles.

*For Examiner's Use*

| particle | number of | | |
|---|---|---|---|
| | electrons | neutrons | protons |
| **A** | 12 | 12 | 12 |
| **B** | 15 | 16 | 15 |
| **C** | 17 | 18 | 17 |
| **D** | 17 | 20 | 17 |
| **E** | 18 | 16 | 16 |
| **F** | 18 | 22 | 18 |
| **G** | 18 | 20 | 20 |

**(a)** What is the nucleon number for **F**?

..... [1]

**(b)** Explain why **A** is a neutral particle.

.....

.....

..... [2]

**(c)** Which particles are isotopes of the same element?

..... [1]

**(d)** What is the charge on **E**?

..... [1]

**(e)** Which particles have the same relative mass?

..... [1]

[Total: 6]

**A5** Analysis of compound **X** shows it has the following composition.

*For Examiner's Use*

| element | percentage by mass |
|---|---|
| nitrogen | 11.1 |
| hydrogen | 3.20 |
| chromium | 41.3 |
| oxygen | 44.4 |

**(a)** Show that **X** has the formula $N_2H_8Cr_2O_7$.

[3]

**(b)** An aqueous solution of **X** is orange.

Suggest which element in **X** is responsible for the orange colour.

..... [1]

**(c)** An acidified aqueous solution of **X** reacts with aqueous potassium iodide to form iodine.

State and explain what you can conclude about the chemical nature of **X**.

.....

.....

..... [2]

**(d)** Aqueous sodium hydroxide is added to solid **X** and the mixture is warmed. A gas that turns moist red litmus blue is evolved.

**(i)** Give the formula of the positive ion present in **X**.

..... [1]

**(ii)** Suggest the formula of the other ion present in **X**.

..... [1]

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**(e)** When solid **X** is heated only $Cr_2O_3$, water and gas **Z** are formed.

*For Examiner's Use*

Name gas **Z**.

..... [1]

[Total: 9]

**Question A6 starts on page 10.**

**A6** Potassium is in Group I and chlorine is in Group VII of the Periodic Table.

*For Examiner's Use*

Potassium forms an oxide with the formula $K_2O$ and chlorine forms an oxide with the formula $Cl_2O$.

**(a) (i)** Draw a 'dot-and-cross' diagram for $Cl_2O$.

You only need to draw the outer shell electrons.

[1]

**(ii)** Explain, using ideas about structure and bonding, why $Cl_2O$ has a low melting point.

.....

.....

.....

..... [2]

**(b)** Draw diagrams to show the electronic structures and charges of both ions present in potassium oxide.

[2]

**(c)** Chlorine forms another oxide $Cl_2O_7$. One mole of this oxide reacts with one mole of water to make two moles of an acid and no other products.

Construct the equation for this reaction.

..... [1]

[Total: 6]

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## Section B

For Examiner's Use

Answer **three** questions from this section in the spaces provided.

The total mark for this section is 30.

**B7** Malachite is an ore of copper. The formula of malachite is $CuCO_3.Cu(OH)_2$.

Malachite reacts as though it is a mixture of copper(II) carbonate and copper(II) hydroxide.

A small sample of malachite is added to excess dilute hydrochloric acid, HC*l*(aq). The carbon dioxide formed is collected and has a volume of 96 $cm^3$ at room temperature and pressure.

**(a)** What would you observe when malachite reacts with HC*l*(aq)?

.....

..... [2]

**(b)** Construct the equation for the reaction between malachite and HC*l*(aq).

..... [2]

**(c)** Calculate the mass of carbonate ion, $CO_3^{2-}$, in the sample of malachite.

mass of $CO_3^{2-}$ = ..... g [3]

**(d)** Copper is extracted from malachite by heating with carbon.

**(i)** Construct an equation for the reduction of malachite by carbon.

..... [2]

**(ii)** Malachite is a finite resource. Give one **other** reason why copper should be recycled.

.....

..... [1]

[Total: 10]

**B8** Carboxylic acids are a homologous series of organic compounds.

*For Examiner's Use*

The table shows information about some carboxylic acids.

| carboxylic acid | formula | melting point / °C | boiling point / °C |
|---|---|---|---|
| methanoic acid | $HCO_2H$ | 8 | 100 |
| ethanoic acid | $CH_3CO_2H$ | 17 | 118 |
| | $C_2H_5CO_2H$ | –22 | 141 |
| butanoic acid | $C_3H_7CO_2H$ | | |
| hexadecanoic acid | $C_{15}H_{31}CO_2H$ | 63 | 269 |

**(a)** What is meant by the term *homologous series*?

.....

.....

.....

..... [2]

**(b)** Name the carboxylic acid with the formula $C_2H_5CO_2H$.

..... [1]

**(c)** Deduce the general formula for a carboxylic acid.

..... [1]

**(d)** It is more difficult to estimate the melting point of butanoic acid than its boiling point. Use the data in the table to explain why.

.....

..... [1]

**(e)** When warmed in the presence of concentrated sulfuric acid, butanoic acid reacts with ethanol to make an ester.

Name and draw the structure, showing all the atoms and all the bonds, of this ester.

name .....

structure

[2]

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**(f)** Hexadecanoic acid, $C_{15}H_{31}CO_2H$, is a weak acid.

*For Examiner's Use*

**(i)** Write an equation to show the dissociation of hexadecanoic acid when dissolved in water. Use the equation to explain the meaning of the term weak acid.

.....

.....

..... [2]

**(ii)** What is the formula of the salt formed when hexadecanoic acid reacts with aqueous sodium hydroxide?

..... [1]

[Total: 10]

**B9** Ethanol is manufactured by the hydration of ethene.

For Examiner's Use

$$C_2H_4(g) + H_2O(g) \rightleftharpoons C_2H_5OH(g) \qquad \Delta H = -45\,kJ/mol$$

This reaction is exothermic.

The reaction is normally carried out at a pressure of 70 atmospheres and a temperature of 300 °C.

**(a)** The reaction is carried out at 70 atmospheres pressure and at **600 °C** rather than 300 °C.

Predict and explain the effect of raising the temperature on

**(i)** the rate of reaction,

.....

.....

..... [2]

**(ii)** the position of equilibrium.

.....

.....

..... [2]

**(b)** The reaction is carried out at **20 atmospheres** rather than 70 atmospheres, and at 300 °C.

Predict and explain the effect of decreasing the pressure on

**(i)** the rate of reaction,

.....

.....

..... [2]

**(ii)** the position of equilibrium.

.....

.....

..... [2]

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**(c)** Calculate the energy released when 10 moles of ethanol are formed.

*For Examiner's Use*

energy released = ..... kJ [1]

**(d)** The hydration of ethene uses an acid catalyst.

Explain how a catalyst can increase the rate of reaction.

.....

..... [1]

[Total: 10]

**B10** Aqueous silver nitrate can be electrolysed using inert electrodes.
Solid silver is formed on the cathode (negative electrode).

*For Examiner's Use*

The table shows how the mass of silver formed is affected by four factors.

| temperature of solution /°C | duration of electrolysis /seconds | current passed through solution/amps | concentration of solution /mol/dm$^3$ | mass of silver formed /g |
|---|---|---|---|---|
| 25 | 100 | 9.65 | 1.0 | 0.108 |
| 30 | 100 | 9.65 | 1.0 | 0.108 |
| 25 | 100 | 9.65 | 0.5 | 0.108 |
| 25 | 200 | 9.65 | 0.5 | 0.216 |
| 25 | 100 | 19.3 | 1.0 | 0.216 |

**(a)** The electrode reaction at the cathode is reduction.

**(i)** Construct the equation for the reaction which occurs at the cathode.

..... [1]

**(ii)** Explain why this reaction is reduction.

.....

..... [1]

**(b)** State how each of the following factors affects the mass of silver formed at the cathode.

temperature of solution

.....

.....

duration of electrolysis

.....

.....

current used

.....

.....

concentration of solution

.....

.....

[4]

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**(c)** Explain why aqueous silver nitrate can be electrolysed but solid silver nitrate cannot.

*For Examiner's Use*

.....

.....

..... [2]

**(d)** Aqueous silver nitrate reacts with dilute hydrochloric acid to form a white precipitate.

Construct the ionic equation, including state symbols, for the formation of this white precipitate.

..... [2]

[Total: 10]

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# DATA SHEET
## The Periodic Table of the Elements

| Group I | Group II | Group III | Group IV | Group V | Group VI | Group VII | Group 0 |
|---|---|---|---|---|---|---|---|
| | | | | | | | 4 He Helium 2 |
| 7 Li Lithium 3 | 9 Be Beryllium 4 | 11 B Boron 5 | 12 C Carbon 6 | 14 N Nitrogen 7 | 16 O Oxygen 8 | 19 F Fluorine 9 | 20 Ne Neon 10 |
| 23 Na Sodium 11 | 24 Mg Magnesium 12 | 27 Al Aluminium 13 | 28 Si Silicon 14 | 31 P Phosphorus 15 | 32 S Sulfur 16 | 35.5 Cl Chlorine 17 | 40 Ar Argon 18 |
| 39 K Potassium 19 | 40 Ca Calcium 20 | 70 Ga Gallium 31 | 73 Ge Germanium 32 | 75 As Arsenic 33 | 79 Se Selenium 34 | 80 Br Bromine 35 | 84 Kr Krypton 36 |
| 85 Rb Rubidium 37 | 88 Sr Strontium 38 | 115 In Indium 49 | 119 Sn Tin 50 | 122 Sb Antimony 51 | 128 Te Tellurium 52 | 127 I Iodine 53 | 131 Xe Xenon 54 |
| 133 Cs Caesium 55 | 137 Ba Barium 56 | 204 Tl Thallium 81 | 207 Pb Lead 82 | 209 Bi Bismuth 83 | 209 Po Polonium 84 | 210 At Astatine 85 | 222 Rn Radon 86 |
| 223 Fr Francium 87 | 226 Ra Radium 88 | | | | | | |

| Transition elements | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 89 Ac Actinium † | 45 Sc Scandium 21 | 48 Ti Titanium 22 | 51 V Vanadium 23 | 52 Cr Chromium 24 | 55 Mn Manganese 25 | 56 Fe Iron 26 | 59 Co Cobalt 27 | 59 Ni Nickel 28 | 64 Cu Copper 29 | 65 Zn Zinc 30 | | | |
| 139 La Lanthanum 57 * | 178 Hf Hafnium 72 | 181 Ta Tantalum 73 | 184 W Tungsten 74 | 186 Re Rhenium 75 | 190 Os Osmium 76 | 192 Ir Iridium 77 | 195 Pt Platinum 78 | 197 Au Gold 79 | 201 Hg Mercury 80 | | | | |

| Lanthanoid series * | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 140 Ce Cerium 58 | 141 Pr Praseodymium 59 | 144 Nd Neodymium 60 | 147 Pm Promethium 61 | 150 Sm Samarium 62 | 152 Eu Europium 63 | 157 Gd Gadolinium 64 | 159 Tb Terbium 65 | 162 Dy Dysprosium 66 | 165 Ho Holmium 67 | 167 Er Erbium 68 | 169 Tm Thulium 69 | 173 Yb Ytterbium 70 | 175 Lu Lutetium 71 |

| Actinoid series † | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 232 Th Thorium 90 | 231 Pa Protactinium 91 | 238 U Uranium 92 | 237 Np Neptunium 93 | 244 Pu Plutonium 94 | 243 Am Americium 95 | 247 Cm Curium 96 | 247 Bk Berkelium 97 | 251 Cf Californium 98 | 252 Es Einsteinium 99 | 257 Fm Fermium 100 | 258 Md Mendelevium 101 | 259 No Nobelium 102 | 260 Lr Lawrencium 103 |

* 58–71 Lanthanoid series
† 90–103 Actinoid series

Key: a = relative atomic mass; X = atomic symbol; b = atomic (proton) number

The volume of one mole of any gas is 24 $dm^3$ at room temperature and pressure (r.t.p.).